

Pankaj Dewangan

AI Engineer

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AI Engineer with 1.7 years of experience building production-grade LLM, RAG, and Agentic AI systems on Azure and GCP. Specialized in LLM system design, cost & latency optimization, enterprise AI integrations, and scalable GenAI platforms. Proven ability to deliver AI solutions used by 1000+ users, reduce operational effort by up to 90%, and achieve 99% availability in production environments.

SKILLS

- GenAI: LLM Engineering Advanced RAG Agentic AI Prompt/Function Calling Guardrails Hallucination Mitigation
- Backend & Data: Python FastAPI (Async) REST WebSockets SQL PostgreSQL Pinecone Redis Oracle
- Architecture: LLM System Design Scalable & Multi-Tenant GenAI Cost/Latency Optimization Hybrid Search (BM25 + Vector) Re-ranking
- Cloud & DevOps: Azure (OpenAI AI Services) GCP (Vertex AI) Docker Kubernetes GitHub/GitLab CI/CD
- LLMOps: MLflow Prompt Versioning LLM Evaluation Model Monitoring CI/CD Canary Deployments
- Frameworks: Copilot Studio LangChain LangGraph

WORK EXPERIENCE

AI Engineer - Capgemini

Oct 2024 - Present

- Designed and handed over end-to-end **GenAI** and **Agentic AI** solutions reducing manual effort by 80% using **Python**, **OpenAI**, **Vertex AI**, **LLaMa**, **LangChain**, and **LangGraph**.
- Built **RAG**-based chatbots and automation solutions with advanced prompt engineering (**zero-shot**, **one-shot**, **few-shot**), improving response accuracy by 30%.
- Established AI solutions with enterprise platforms (**ServiceNow**, **Oracle DB**, **Power BI**, **Microsoft Teams**, **Google Workspace**), enabling seamless workflows for **1,000+** users.
- Achieved **scalable data storage** and retrieval using **PostgreSQL**, **Azure Cosmos DB**, and **Pinecone**, supporting **millions of embeddings** with **sub-second latency**.
- Delivered **production-grade AI applications** on Azure and GCP, achieving **99.9% availability** and reducing infrastructure costs by **25%**.
- Established **CI/CD pipelines** with **GitLab** and **GitHub**, **accelerating release cycles by 50%** and improving deployment reliability.

PROJECTS

Interactive Voice Response (IVR) – GenAI Voice Bot

2 Months

- Architected an **enterprise-grade GenAI voice recognition system** enabling real-time multilingual (**3 languages**) speech processing for customer service.
- Implemented **RAG** with **vector databases**, improving response relevance and accuracy by **~35%**.
- Optimized **WebSocket-based streaming APIs**, reducing end-to-end latency from **~1.8s to ~1.0s**.
- Enabled **real-time order status** and store-specific responses, improving engagement & **operational efficiency** by **~25%**.

Tech Stack: Azure AI, Azure Speech Services, Python, Vector DB, RAG, WebSockets, OpenAI GPT-4.

Assisted Ticket Resolution – Gen AI Solutions4 Months

- Developed an **LLM-powered ticket resolution** system leveraging historical incident data to automatically suggest the **top 3 relevant solutions** for new tickets using semantic similarity.
- **Reduced Mean Time to Resolve (MTTR)** by up to **80%** by enabling faster incident resolution through context-aware recommendations.
- Implemented **GenAI-based response generation** for new/unseen tickets using trained LLM models, improving resolution accuracy and support efficiency.
- Created the solution on Microsoft Azure using **Azure Bot Service** and FastAPI, improving scalability and reliability.

Tech Stack :- Python, OpenAI(LLM), Embeddings, Vector Database, Semantic Search

Financial Analysis and Competitor Benchmarking Tool4 Months

- Developed a **GenAI-driven financial analysis tool** to evaluate competitor performance using **balance sheets and income statements**, enabling data-driven benchmarking and insights.
- Applied **financial ratio analysis (ROA, ROE, profit margins)** and variance analysis to identify key **areas of improvement**, helping clients optimize strategies and improve profitability.
- Automated competitor analysis and insight **generation using LLMs**, reducing manual analysis effort by **70%+** and accelerating decision-making.

Tech Stack :- Python, OpenAI, Vertex AI, Pandas, Financial Modeling, Data Analysis

GenAI-Powered Chatbot & Voice Assistant with Power BI Integration:2 Months

- Positioned a **voice and query-based AI chatbot** unified with **Power BI** dashboards, enabling **real-time insights and notification** from business data.
- Crafted efficient dataset querying mechanisms, elevating data accuracy by 25% and truncating query time by ~40%, accelerating overall insight delivery.
- Leveraged **GenAI** and **Azure Speech Services** to enable natural language and voice interactions, increasing dashboard adoption by ~30%.
- **Improved decision-making speed** and **operational effectiveness** by delivering instant, conversational access to key metrics.

Tech Stack: Python, Pandas, GenAI, Azure Speech Services, Power BI

EDUCATION

M. Tech Computer Science & Engineering Motilal Nehru National institute of Technology Allahabad (NIT ALLAHABAD)	2022 - 2024 CPI – 7.47/10
B. Tech Computer Science & Engineering Christian College of Engineering and Technology Bhilai (CCET BHILAI)	2022 - 2024 CPI – 9.05/10